r/NoStupidQuestions

How does AI use water for cooling? Wouldn't that just fry the electronics?

Posted by u/HeiressOfMadrigal • • 0 points • 14 comments

I can't cool down my computer by pouring a glass of water on top of it. I have no doubt that AI is bad for the environment and uses water, but how exactly does that work?

EDIT: Thank you for all the great answers! I understand now. I figured it was something like that but was just curious 🙂

Comments

u/ntengineer • 1 points • 2026-03-27 06:26 UTC

Same way that high end gaming systems have water cooling. The water is kept away from sensitive parts

u/navelencounters • 1 points • 2026-03-26 22:22 UTC

I work for companies that design & build data centers...the water is just used for temperature control like the air conditioning in your home...the water never touches the components...the water is cooled, then fans blow air over the cold pipes...the hot air gets exhausted...it just takes a LOT of water all the time!...you can only imagine the safety protocals we have engineered in to prevent over heating, especially if we lose power...our window is 3 minutes, beyond that time the heat will rise and destroy most components.

u/DuckDuckGo-8857 • 1 points • 2026-03-26 20:13 UTC

If quantum computers is on the cusp of being commercially available, why hasn't these companies with their bottomless resources expedite the technology and drastically reduce their demand on resources and at the same time exponentially increasing their productivity?

When I hear about the astronomical resources these AI farms require, I think of the movie Transcendence where a whole field is occupied by computers.

> **u/Linkpharm2** • 1 points • 2026-03-26 22:10 UTC
>
> Because developers have spent the last 50 years making computers do things in really smart ways, and that would need to be redone. So basically software is the issue.

u/NDaveT • 2 points • 2026-03-26 19:44 UTC

> I can't cool down my computer by pouring a glass of water on top of it.

If you have a friend who built a fancy gaming PC, they might have used liquid cooling for the CPU, GPU, or both. There are tubes filled with water and/or antifreeze that run from the processors to a heat exchanger. Computers in datacenters use similar cooling but just scaled way up.

u/MegaBusKillsPeople • 0 points • 2026-03-26 19:44 UTC

The same manner as in your car.

u/East-Bike4808 • 1 points • 2026-03-26 20:00 UTC

My car doesn’t evaporate water to stay cold.

u/MegaBusKillsPeople • 1 points • 2026-03-26 20:03 UTC

It does to a point, but most modern data centers use closed loop systems.

u/East-Bike4808 • 2 points • 2026-03-26 20:06 UTC

They do for the inner loop, but the outer loop is almost always still either chillers (big AC) or cooling towers (water-evaporators). Heat exchangers cool the inner loop with the outer/tower loop.

Source: I'm in a modern datacenter right now, and it's the latter kind. We're slim on electricity for plant cooling, but we guzzle a lot of water to do it.

u/MegaBusKillsPeople • 1 points • 2026-03-26 20:08 UTC

The chillers would have ver little waste. The evaporators would lose a large amount to the atmosphere. I the actual amount I guess would be dependent on relative humidity and temperature.

u/East-Bike4808 • 1 points • 2026-03-26 20:09 UTC

Right: the chillers don't use water (it's not water). The cooling towers make clouds :-). High water use isn't as much of a limiting factor for many datacenters like power availability is, so e.g. in our case it's the lesser of two evils to do the job.

u/MegaBusKillsPeople • 1 points • 2026-03-26 20:11 UTC

Chillers are likely using glycol on one side and compressed refrigerant another side. I'd imagine it's the same system they use to chill beer lines in a tap system just on a really large scale.

u/DiggingInGarbage • 6 points • 2026-03-26 19:39 UTC

Water cooling is done by having the water in tubes, which run next to the parts of the computer that need to be cooled, heat is conducted into the water since it’s colder, then the water is pumped away

u/Astramancer_ • 5 points • 2026-03-26 19:39 UTC

Ever hear the term "swamp cooler"? (aka evaporative cooling)

That's how, they run air conditioners and cool the hot side of the air conditioner with a spray of water (to simplify the explanation) that evaporates away sucking up huge amounts of heat compared to just blowing ambient air across it. It saves a significant amount of electricity at the cost of water.

The water doesn't get anywhere near the electronics.